GRADE 10

Memo-Questions from a past Maths cycle test

QUESTION 2

A function $f(x)$ is defined as: $f(x)=7x^2-x+6$

(a) Determine $f(-2)$. (2)

$f(-2)=7(-2)^2-(-2)+6$

$f(-2)=36$

(b) Find x if $f(x)=12$. (3)

$12=7x^2-x+6$

$0=7x^2-x-6$

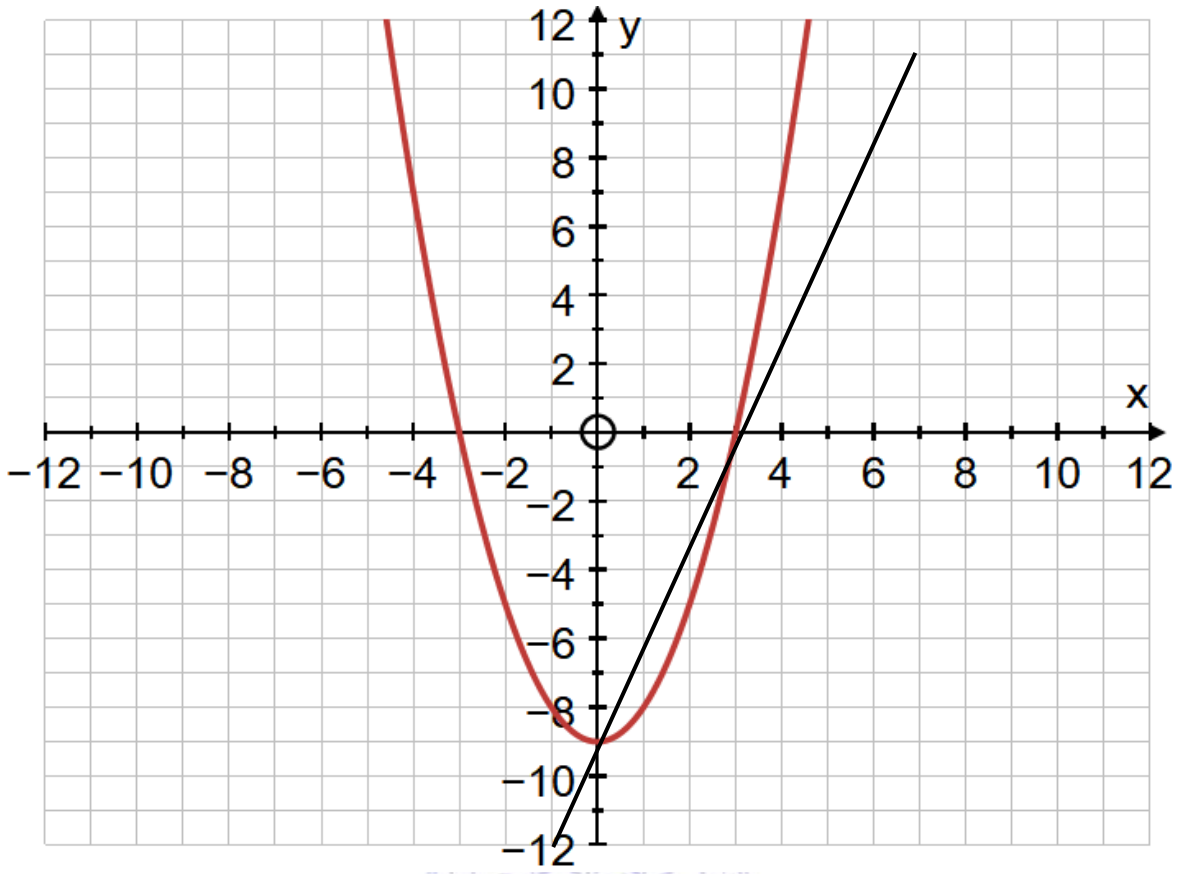
$0=(x-1)(7x+6)$

$x_1=1 \text{ OR } x_2=-\frac{6}{7}$

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QUESTION 4

Given: $f(x)=x^2-9$



(a) On the same set of axis sketch $g(x)=3x-9$, clearly show all intercepts. (2)

(b) Write down the range of f. (2)

$y\in[-9;\infty)$

(c) For what values of x is $f(x)\geq 0$? (2)

$x\in(-\infty;-3]\cup[3;\infty)$

(d) Give the coordinates of the points of intersection between $f(x)$ and $g(x)$. (4)

$(0;-9)$

$(3;0)$

(e) If f is moved up 5 units, what is the equation of the new function? (1)

$f(x)=x^2-4$

(f) For which values of x is $f(x)$ decreasing? (2)

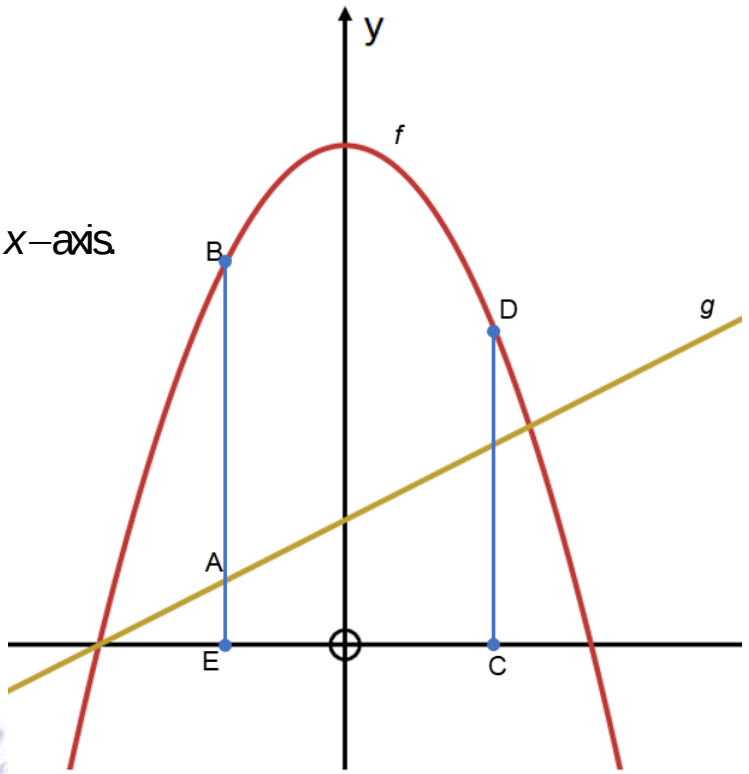
$x\in(-\infty;0)$

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QUESTION 5

In the diagram alongside:

- $f(x)=16-x^2$
- $g(x)=x+4$
- Lines EAB and CD are perpendicular to the $x-$axis.



(a) If OE is 2 units, calculate the lengths of:

(1) AE (2)

$AE=2$

(2) BE (1)

$BE=12$

(3) AB (2)

$BE-AE=12-2$

$AB=10$

(b) If CD is 7 units, calculate the length of OC. (3)

$D(3;7)$

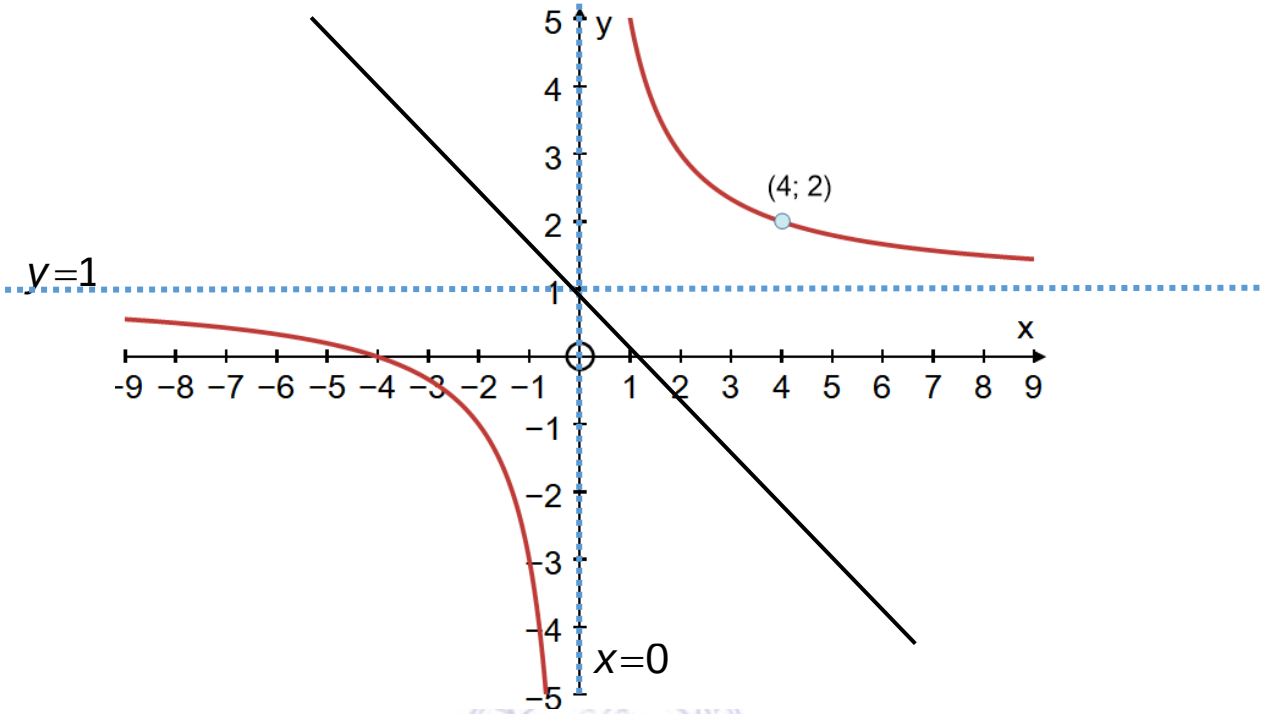
$C(3;0)$

$OC=3$

[8]

QUESTION 6

The equation of the function $g(x)=\frac{a}{x}+q$ is shown below. It passes through the point $(4;2)$ on the graph of g and has a range of $y\in(-\infty;1)\cup(1;\infty)$.



(a) On the above diagram, clearly indicate both asymptotes of $g(x)$. (2)

(b) Determine the equation of g. (3)

$g(x)=\frac{a}{x}+1$

$2=\frac{a}{4}+1$

$a=4 \qquad g(x)=\frac{4}{x}+1$

(c) Determine the equation of h, the axis of symmetry of g which has a negative gradient. (2)

$y=-x+1$

(d) Sketch the graph of h on the above diagram. (2)

(e) Determine the equation of $p(x)$, which is the reflection of $g(x)$ about the $y-$axis. (2)

$p(x)=-\frac{4}{x}+1$

(f) Write the equations of the asymptotes of $f(x)$ if $f(x)=g(x)+3$. (2)

$x=0$

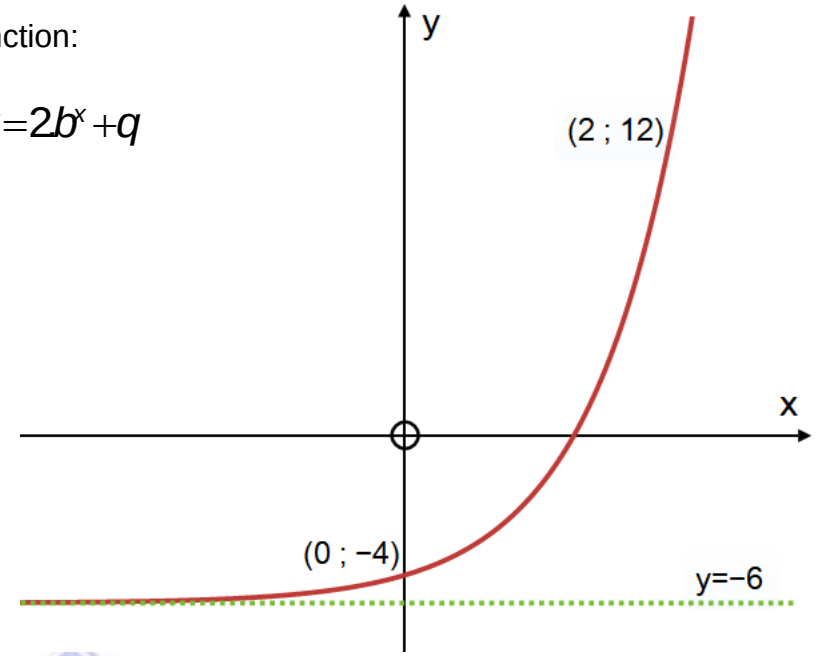
$y=4$

[13]

QUESTION 7

Determine the equation of the following function:

$$y=2.b^x+q$$



$y=2.b^x-6$
$Sub(2;12)$
$12=2.b^2-6$
$b=\pm 3$
$b=3$ $\quad y=2.3^x-6$

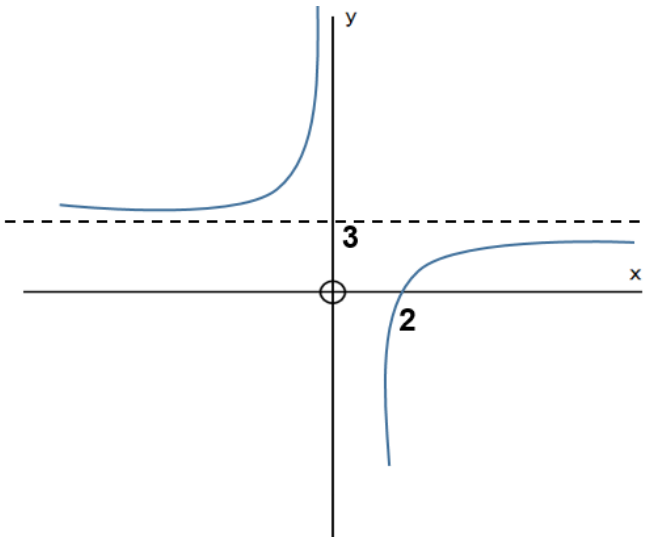
[4]

QUESTION 8

A hyperbola, h, is described with the following characteristics:

- The equation of the vertical asymptote is $x=0$
- The range of h is $y\in\mathbb{R}, y\neq 3$
- The x-intercept of h is $(2;0)$

Draw a rough sketch of h. Show all asymptotes and intercepts.

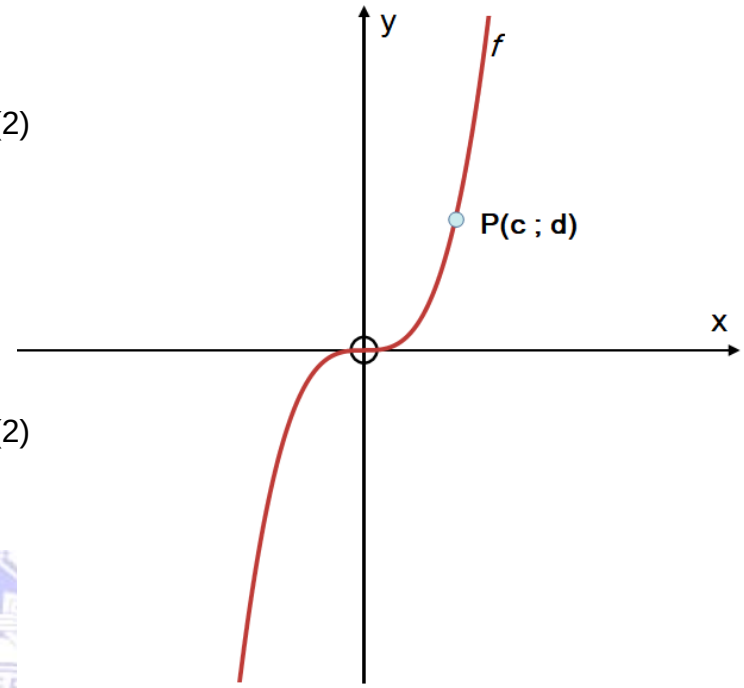


[4]

QUESTION 9

Sketched is $y=f(x)$ and $P(c;d)$.

Give the coordinates of ***P*** in terms of ***c*** and ***d*** if:



(a) $y=2f(x)$ (2)

$(c;2d)$

(b) $y=f(-x)+2$ (2)

$(-c;d+2)$

[4]